

Solution

LIGHT-02

Class 08 - Science

Section A

1. Fill in the blanks:

- (i) 1. thinner
- (ii) 1. Bright
- (iii) 1. Virtual
- (iv) 1. Large
- (v) 1. Iris
- (vi) 1. Lesser (fewer)
- (vii) 1. Dispersion
- (viii) 1. 63
- (ix) 1. Multiple Reflection
- (x) 1. Regular
- (xi) 1. A
- (xii) 1. Left, Left hand
- (xiii) 1. Directly
- (xiv) 1. Diffused
- (xv) 1. 2
- (xvi) 1. Irregular
- (xvii) 1. Retina
- (xviii) 1. Equal

Section B

2. (a) Do not require medium

Explanation: Light doesn't need a medium through which to travel because the speed of light is experimentally constant: independent of the movement of the source or detector or the direction in which it travels.

3.

(b) 63

Explanation: Braille system has 63 dot pattern that can be felt with finger tips to understand the matter written on it.

4.

(c) 16

Explanation: Human eye can see maximum 16 separate pictures in a second. The images formed on retina remain up to 1/16th second. This phenomenon is called persistence of vision.

Section C

5. (a) Cones are sensitive to bright light

Explanation: Cones are sensitive to bright light and they sense colour. However, rods are sensitive to dim light.

6.

(b) Always

Explanation: First law of reflection: When a light ray strikes a plane mirror, the light ray reflects off the mirror. Reflection involves a change in the direction of the light ray. The angle of incidence is the angle between this normal line and the incident ray; the angle of reflection is the angle between this normal line and the reflected ray. According to the law of reflection, the angle of incidence equals the angle of reflection.

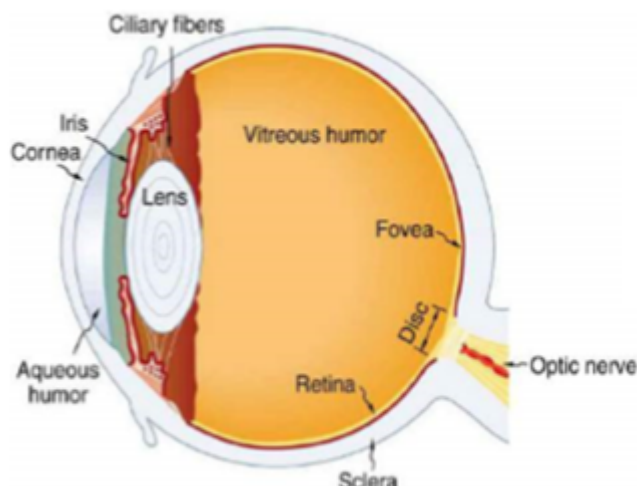
Section D

7. The eyeball is covered with a tough layer. This layer is transparent on the front. This transparent portion is called cornea.

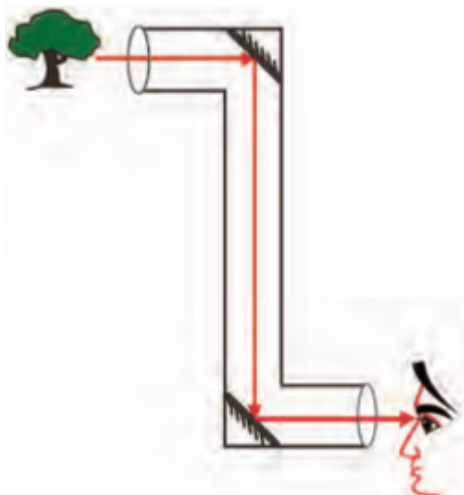
8. When an image is formed by a mirror the left of the object appears on the right and the right appears on the left, this is known as lateral inversion.

9. Two (cones and rods).
 10. Rainbow is a natural phenomenon showing dispersion.
 11. Louis Braille, himself a visually challenged person, developed a system for visually challenged persons and published it in 1821.
 12. There is a hole in the centre of iris. This is called pupil.
 13. Rods are the cells which are sensitive to dim light.
 14. The ray that comes back from the surface after reflection is known as the reflected ray.
 15. For one incident ray, there is one and only reflected ray.
 16. Cones are the cells which are sensitive to bright light.
 17. The ray which comes from object and strikes a surface is called as incident ray.
 18. When light falls on polished surfaces, they change the direction of light in a well-defined manner. This is called regular reflection.
 19. The iris increase and decrease the size of pupil and regulate the amount of light that enters through the pupil.
 20. When the mirrors are parallel to each other, then we get infinite number of images.
 21. Iris is colored part of the eye .
 22. Pupil.
 23. The eyelids act as the shutters of the eye and protect them from injury.
 24. Light travels in a straight line. This is called rectilinear propagation of light.
 25. The angle between the normal and incident ray is called the angle of incidence.
 26. When all the reflected rays are parallel to each other; this case of reflection is called regular reflection.
 27. Images are formed on the retina and thus it works like a screen.
 28. Normal is a line making an angle of 90 degree to the line representing the mirror at the point where the incident ray strikes the mirror.
 29. The angle between the normal and reflected ray is called angle of reflection.
 30. Splitting of light into its seven colours is known as dispersion of light.
 31. When two mirrors are kept parallel to each other then numerous images of an object are seen in these mirrors. This is known as multiple images.
 32. At the junction of the optic nerve and retina, there are no sensory cells, so no vision possible at the spot. This is called the blind spot.
 33. Phenomenon of changing left to right and right to left by the mirror while forming images is called lateral inversion.
 34. Cones are sensitive to bright light and colour whereas rods are sensitive to dim light.
 35. When all the parallel rays reflected from a plane surface are not parallel, the reflection is known as diffused or irregular reflection.
 36. Nerve cells.
 37. Designers of wallpapers, fabrics, artists use kaleidoscope to get ideas for new patterns.
 38. The most comfortable distance at which one can read with normal eyes is about 25 cm.
 39.
 - i. Iris
 - ii. Pupil
 - iii. Iris
 - iv. Optic nerve
 40. Image of an object retain on retina for $\frac{1}{16}$ th of a second even after removed is called persistence of vision. This form the basis of the principle of cinematography.
In moving cinema, we take still pictures. The sequence of still pictures are projected on screen at rate of 24 images per second (persistence of vision is $\frac{1}{16}$ th of second.) The successive images merge smoothly and give us feeling of movie.
 41. The impression of an image persists for about 1/16th of a second on the retina. This is known as persistence of vision. If still images of a moving object are flashed on the eye at a rate faster than 16 per second, then the eye perceives this object as moving. So, in the case of waving a hand very fast in front of eyes, the rate of movement of the hand becomes very large (much faster than 16 per second), therefore, the fingers appear blurred.
 42. It is a long tubular device through which a person can see objects that are out of the direct line of sight.
There are two plane mirrors in the periscope.
- Uses of Periscope:**
- i. A Periscope is used to see over the heads of crowd. (say, as in a football match)
 - ii. A Periscope is used by soldiers sitting in a trench (or bunker) to observe the enemy activities outside (over ground).
 - iii. A Periscope is used by a navy officer sitting in a submarine to see ships over the surface of water in the sea.
43. Eye is the organ of vision and acts as a vital optical instrument. It is spherical in shape of about 2.3 cm diameter .

- i. Sclera: It externally covered by a white opaque layer called Sclerotic (sclera) to protect and holds the eye.
- ii. Choroid: The choroid is the vascular layer of the eye lying between the retina and the sclera. The choroid supplies blood to nourish the outer layers of the retina.
- iii. Cornea: It is the transparent part, bulging out at front of eye and allows the light to enter in the eye. Cornea is also known as the window or aperture of the eye.
- iv. Iris and pupil: The coloured ring like structure behind cornea is called iris. It has pupil through which light reach to eye lens.
- v. Eye lens and ciliary muscles: The eye lens is a crystalline convex lens made up of transparent and flexible tissues. It is held by ciliary muscles behind the pupil. Ciliary muscles change the focal length of eye lens.
- vi. Retina: Retina acts as a screen to obtain the image of the object. It contains number of sensitive cells called rods and cones.
- vii. Photoreceptors: Rods detect light and cones detect colour: They first convert light into chemical energy, and then into electrical energy to send it through the optic nerve to the brain.
- viii. Fluid: The aqueous humor is watery fluid which fills the space between cornea and lens. The space between eye lens and retina is filled with another jelly like fluid called vitreous humour.



44. **Mirror Periscope:** The working of a periscope is based on the principle of successive reflections from two plane mirrors. It consists of two plane mirrors facing each other fixed at 45 degree angle to the frame work of a tube. Fix the two mirrors at an angle of 45 degree as shown in the figure. View object through one end.

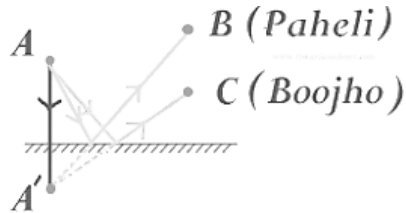


45. i. No. of images = $\frac{360}{90} - 1 = 4 - 1 = 3$
 ii. No. of images = $\frac{360}{60} - 1 = 6 - 1 = 5$
46. i. The eyes should be removed within 4-6 hours after death.
 ii. People infected with AIDS, Hepatitis B and C, rabies, tetanus, etc cannot donate their eyes.
 iii. It is because in bright sunlight the size of the pupil of our eye is small. So as we enter the darkened cinema hall very little light enter our eye and we cannot see things properly. Later the pupil of our eye expands and becomes large and more light enters our eye and we can see clearly.
47. The ability of the eye lens to adjust its focal length to see far and nearby object is called accommodation. The eye lens is a crystalline convex lens made up of transparent and flexible tissues behind the pupil and held by the muscles called ciliary muscles. Eye lens focus inverted the images of objects on the retina of the eye.

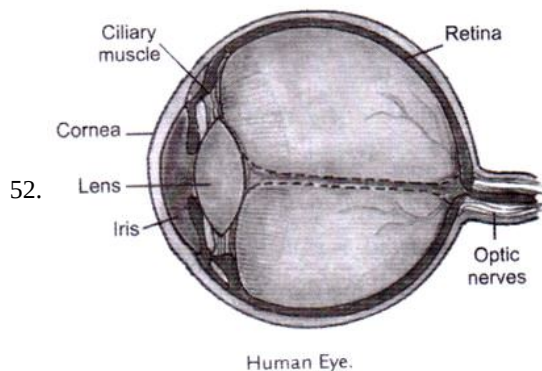
When we are looking at nearby objects, the ciliary muscles contract to increase thickness of eye lens. This increases the curvature of the eye lens therefore the focal length of the eye lens decreases. This enables us to see nearby objects clearly.

When we see far objects, the ciliary muscles relax to decrease thickness of eye lens (lens becomes thin). Thus, its focal length increases. This enables us to see distant objects clearly.

48. The human eye focusses the image for different objects at different distance by changing the focal length of the lens. This is done by the ciliary muscles, which stretch and relax to change the focal length of the lens. This action of the eye is called the power of accommodation of the eye. The most comfortable distance of the normal eye can read is about 25cm. This distance is called the least distance of the eye. The minimum distance at which the eye can see objects distinctly varies with age.
49. i. The image of the object at position A will be formed behind the mirror. It will be at the same distance away from the mirror as the object is B.
 ii. Yes, Paheli at B can see the object because a reflected ray from A will reach B.
 iii. Boojho can also see the image because his eyes receive the reflected ray from A.
 iv. If we trace the reflected rays from B and C backward, they converge at a point behind the mirror. The position of image A remains fixed even if Paheli moves from B to C.



50. i. choroid
 ii. Sclera
 iii. Aqueous humour
 iv. Yellow spot
51. When the light coming from an object enters our eyes, then we can able to see object.
 a. We cannot see the book behind a wooden screen because wood is a non luminous object and cannot emit their own light to make book visible.
 b. We cannot see it in a dark room because there is no source of light in a dark room so to see an object we need a source of light.

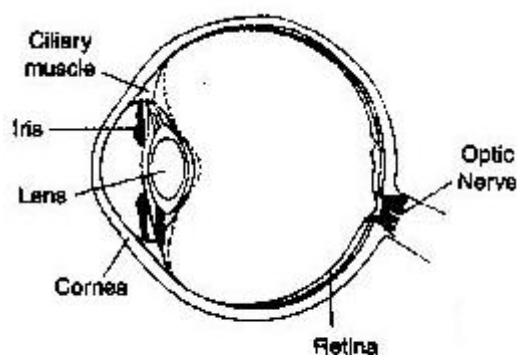


The eye has a roughly spherical shape. The outer coat of the eye is white. It is tough so that it can protect the interior of the eye from accident. Its transparent front part is called cornea. Behind the cornea, we find a dark muscular structure called iris. In the iris, there is a small opening called pupil. The size of the pupil is controlled by the iris. The iris controls the amount of light entering into the eye. Behind the pupil of the eye is a lens which is thicker at the centre. The lens focuses light on the retina. The retina contains several nerve cells. Sensations felt by the nerve cells are then transmitted to the brain through the optic nerves. There are two kinds of cells; cones and rods.

53. i. The image formed by a plane mirror is virtual.
 ii. The image formed by a plane mirror is behind the mirror.
 iii. The image formed by a plane mirror is at same distance behind the mirror as the object is in front of it.
 iv. The image formed by a plane mirror is of the same size as the object.
 v. The image formed by a plane mirror is erect.
 vi. The image formed by a plane mirror is laterally inverted.
54. The eye has roughly spherical surface. The outer coat of eyes is white and tough to protect the interior of eyes from any kind of accident. Its transparent front part is called cornea, behind the cornea there is a dark muscular structure called iris. In the iris there

is small opening called pupil. The size of the pupil is controlled by the iris, the iris controls the amount of light entering into the eye.

55. i. **Pupil**- The inner aperture that we can see in the centre of the eye is known as pupil. It is like a hole through which light enters the eye.
- ii. **Iris**- It is coloured part of the eye and is involved mainly in controlling the size of pupil. The iris increase and decrease the size of pupil and regulate the amount of light that enters through the pupil.
- iii. **Sclera**- The white part of the eye that we can see is known as sclera. It is filled with a clear watery fluid.
- iv. **Cornea**- It is a thin transparent tissue that covers the front of the eye.
- v. **Retina**- Just behind the eyeball, there is a lining called the retina. It is retina that is sensitive to light and has receptors called rods and cones. The rods and cones respond to light and generate impulses.
- vi. Rods are the cells which are sensitive to dim light.
- vii. Cones are the cells which are sensitive to bright light.
- viii. **Ciliary Muscles**- They controls the focal length of the eye lens.
- ix. **Lens**- It is the transparent tissue between pupil and retina. The lens helps in focusing the light that passes through the pupil into eye. This helps in focusing the image on the retina, by bending the light rays.
- x. **Optic Nerve**- It connects the eye to brain and carries impulse to and from the brain.
- xi. **Blind Spot**- The portion on retina where the nerves fibres enters the optic nerve. This portion does not have any rods and cones. The image falling on this portion of retina cannot be seen. This spot is known as Blind Spot.



Human eye

56. Regular reflection	Diffused reflection
1. It takes place from a smooth or regular surface 2. Here all reflected rays are parallel to each other 3. Reflected rays go in one direction.	1. It takes place from a rough surface 2. Here all reflected rays are not parallel to each other 3. Reflected rays are scattered in different direction.

In diffused reflection each ray obeys the law of reflection, thus there is no failure of law of reflection.